MILITARY STANDARD

INSPECTION PROCEDURE FOR DETERMINING THE MAGNETIC PERMEABILITY OF WROUGHT AUSTENITIC STEELS

TO ALL ACTIVITIES:

1. THE FOLLOWING PAGES OF THIS STANDARD HAVE BEEN REVISED AND SUPERSEDE THE PAGES LISTED:

NEW PAGE	DATE	SUPERSEDED PAGE	DATE
1	26 July 1956	(Reprinted without change)	
2	29 November 1957	2	26 July 1956

2. RETAIN THIS COVER PAGE AND INSERT BEFORE THE TABLE OF CONTENTS OF THIS STANDARD.

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MILITARY STANDARD

INSPECTION PROCEDURE FOR DETERMINING THE MAGNETIC PERMEABILITY OF WROUGHT AUSTENITIC STEELS

1. SCOPE

1.1 This standard covers the inspection procedure which shall be used for determining the magnetic permeability of wrought austenitic steel applicable to purchases where essentially nonmagnetic material is specified. This standard shall not be used for material intended for periscope tubes or accessories.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, of the issue in effect on date of invitation for bids, form a part of this standard:

SPECIFICATIONS

FEDERAL

QQ–M–151 — Metals, General Specifications for Inspection of.

QQ–S–763 — Steel Bars, Shapes, and Forgings — Corrosion Resisting.

QQ–S–766 — Steel, Corrosion Resisting: Plates, Sheets and Strips.

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MIL–S–854 — Steel, Corrosion-Resisting: Plates, Sheets, Strips, and Structural Shapes.

MIL–I–17214 — Indicator, Permeability; Low MU (Go-No-Go).

MIL–S–17758 — Steel Bar and Steel Forging, Nickel Modified Hadfield and Nickel-Molybdenum Modified Hadfield.

MIL–S–17759 — Steel Bar, Billet, Forging and Wire, Chromium-Nickel-Phosphorous Alloy, Low Magnetic Permeability.

MIL–S–17996 — Steel Sheet and Other Wrought Forms, Corrosion Resisting, Austenitic (Low Magnetic Permeability), Chromium-Manganese-Nickel.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 The inspection procedures detailed herein are applicable to the austenitic steels. Separate procedures are specified, depending upon the class of materials (see 3.1.2.1).

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3.1.2 *Classes of materials.*

3.1.2.1 For inspection purposes wrought austenitic steels are divided into the following classes:

Class 1 — Those analyses that are not known to exhibit significant tendencies toward ferrite segregation.

Class 2 — Those analyses that are known to exhibit significant tendencies toward ferrite segregation.

3.1.2.2 *Class 1 material.* Class 1 material includes the compositions that conform to the following AISI standard steel types, specifications, or descriptions:

AISI–302
AISI–303
AISI–304
AISI–305
AISI–310

MIL–S–854
Class 1—condition A
Class 2—condition A
Class 11—condition A

QQ–S–763
Class FS–302, condition A
Class FS–303, condition A
Class FS–304, condition A
Class FS–310, condition A

QQ–S–766
Class FS–302—condition A
Class FS–304—condition A
Class FS–310—condition A

3.1.2.2.1 All manganese type austenitic steels that depend primarily on manganese for austenite stability but which may also contain various amounts of nickel, chromium, and molybdenum, in addition to the elements ordinarily found in carbon steel, shall be in class 1.

MIL–S–17758
Alloy 1
Alloy 2

MIL–S–17996
Class 1
Class 2

3.1.2.3 *Class 2 material.* Class 2 material includes compositions conforming to the following AISI standard steel types, specifications, or descriptions:

AISI–321
AISI–347
AISI–304L
AISI–316
AISI–316L

MIL–S–854
Class 8
Class 9

QQ–S–763
Class FS–304L
Class FS–316
Class FS–316L
Class FS–321
Class FS–347

QQ–S–766, condition A of the following classes:
Class FS–321
Class FS–347
Class FS–304L
Class FS–316
Class FS–316L

MIL–S–17759

3.1.3 Austenitic steels that are not included in class 1 or 2 shall be considered class 2 material.

3.1.4 *Permeability.* The permeability of classes 1 and 2 shall not exceed 1.2.

3.2 Inspection equipment.

3.2.1 A permeability indicator conforming to Specification MIL–I–17214 shall be used.

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